

matrixdiagrams.sty

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1 Main command

`\matrixdiagram[scale][aspect]{content}`

1.1 Scale (aspect = 1)

$\begin{bmatrix} & \end{bmatrix}$	<code>\matrixdiagram[1]{\mdBlank}</code>	$\begin{bmatrix} & \end{bmatrix}$	<code>\matrixdiagram[2]{\mdBlank}</code>
$\begin{bmatrix} & \end{bmatrix}$	<code>\matrixdiagram[3]{\mdBlank}</code>	$\begin{bmatrix} & \end{bmatrix}$	<code>\matrixdiagram[4]{\mdBlank}</code>
$\begin{bmatrix} & \end{bmatrix}$	<code>\matrixdiagram[5]{\mdBlank}</code>		

1.2 Aspect ratio (scale = 3)

$\left[\begin{array}{c} \\ \\ \end{array} \right]$	<code>\matrixdiagram[3][0.5]{\mdBlank}</code>	$\left[\begin{array}{c} \\ \\ \end{array} \right]$	<code>\matrixdiagram[3][1]{\mdBlank}</code>
$\left[\begin{array}{c} \\ \\ \end{array} \right]$	<code>\matrixdiagram[3][1.5]{\mdBlank}</code>	$\left[\begin{array}{c} \\ \\ \end{array} \right]$	<code>\matrixdiagram[3][2]{\mdBlank}</code>
$\left[\begin{array}{c} \\ \\ \end{array} \right]$	<code>\matrixdiagram[3][3]{\mdBlank}</code>		

2 Diagonal commands

2.1 `\mdDiag` — main diagonal

$\left[\begin{array}{c} \\ \\ \end{array} \right]$	$\left[\begin{array}{c} \\ \\ \end{array} \right]$	<code>\matrixdiagram[3]{\mdDiag}</code>
		<code>\matrixdiagram[3][2]{\mdDiag}</code>

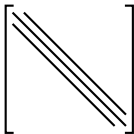
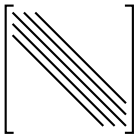
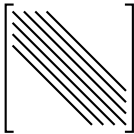
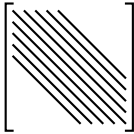
2.2 `\mdDiagBand` — tridiagonal band

Three parallel anti-diagonals with spacing $d = 0.1$.

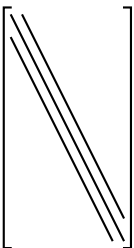
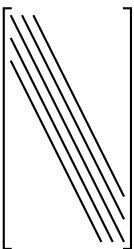
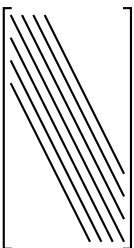
$\left[\begin{array}{c} \\ \\ \end{array} \right]$	$\left[\begin{array}{c} \\ \\ \end{array} \right]$	<code>\matrixdiagram[3]{\mdDiagBand}</code>
		<code>\matrixdiagram[3][2]{\mdDiagBand}</code>

2.3 `\mdDiagBandN{n}` — variable band width

$n = 3, 5, 7, 9$ (spacing fixed at $d = 0.1$ per diagonal):

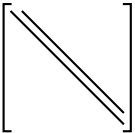
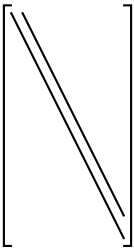
	<code>\matrixdiagram[3]{\mdDiagBandN{3}}</code>
	<code>\matrixdiagram[3]{\mdDiagBandN{5}}</code>
	<code>\matrixdiagram[3]{\mdDiagBandN{7}}</code>
	<code>\matrixdiagram[3]{\mdDiagBandN{9}}</code>

Note: `\mdDiagBandN{3}` is identical to `\mdDiagBand`. With `aspect = 2`:

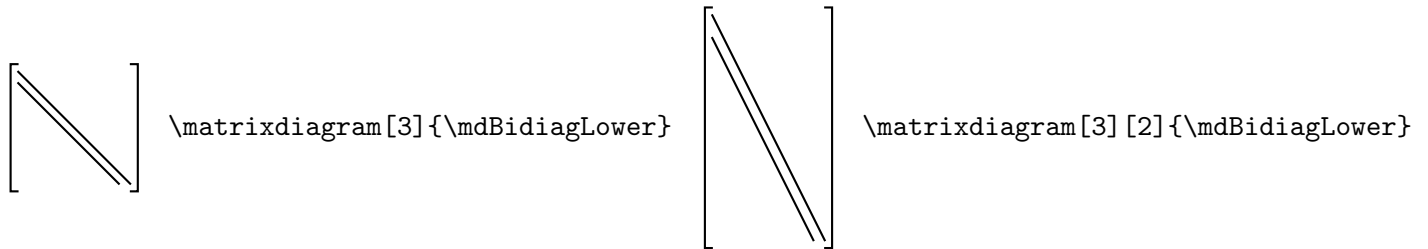
	<code>\matrixdiagram[3][2]{\mdDiagBandN{3}}</code>
	<code>\matrixdiagram[3][2]{\mdDiagBandN{5}}</code>
	<code>\matrixdiagram[3][2]{\mdDiagBandN{7}}</code>

3 Bidiagonal commands

3.1 `\mdBidiagUpper` — main + super-diagonal

	<code>\matrixdiagram[3]{\mdBidiagUpper}</code>		<code>\matrixdiagram[3][2]{\mdBidiagUpper}</code>
---	--	---	---

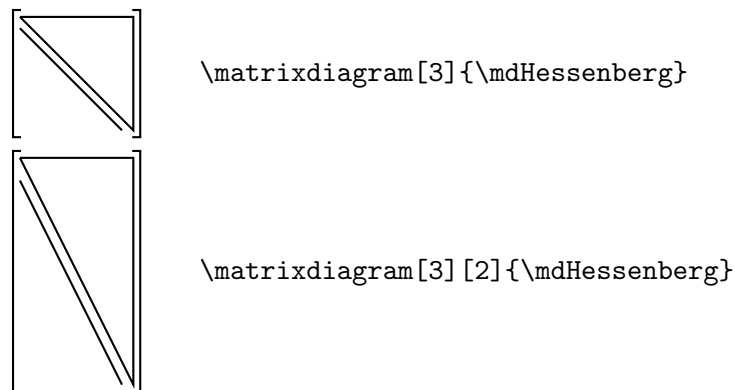
3.2 `\mdBidiagLower` — main + sub-diagonal



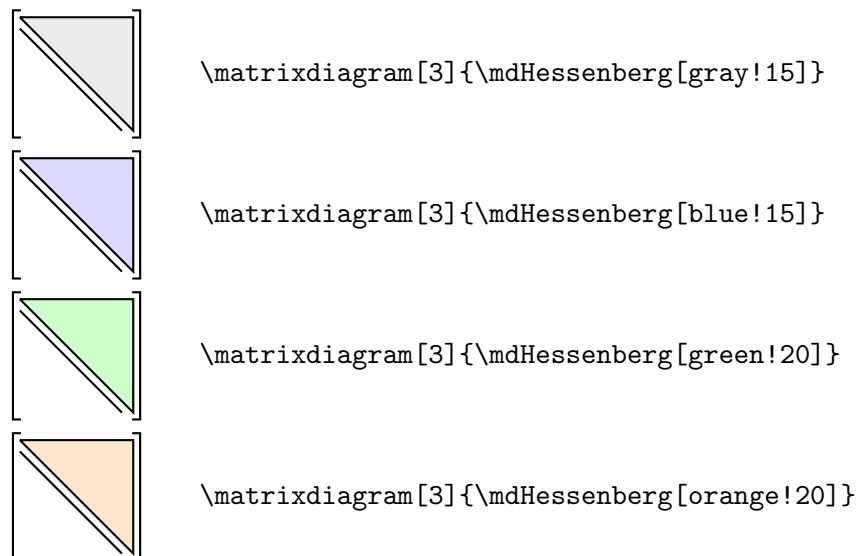
4 Hessenberg command

4.1 `\mdHessenberg` — outline only (default)

Upper-triangular outline $(0, H) \rightarrow (1, H) \rightarrow (1, 0)$ plus one sub-diagonal.



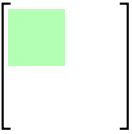
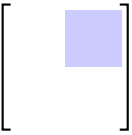
4.2 `\mdHessenberg[color]` — with fill



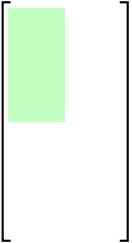
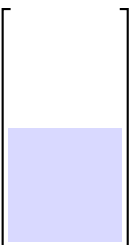
5 Fill and line commands

5.1 `\mdFill[color]{x0}{y0}{x1}{y1}`

Four quadrant fills; default color is `gray!20`:

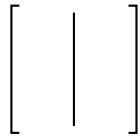
	<code>\matrixdiagram[3]{\mdFill[green!30]{0}{0}{0.5}{1}}</code>
	<code>\matrixdiagram[3]{\mdFill[blue!20]{0.5}{0.5}{1}{1}}</code>
	<code>\matrixdiagram[3]{\mdFill[red!20]{0}{0}{0.5}{0.5}}</code>
	<code>\matrixdiagram[3]{\mdFill[orange!25]{0.5}{0}{1}{0.5}}</code>
	<code>\matrixdiagram[3]{\mdFill{0}{0}{1}{1}}</code>

With `aspect = 2` (y -coordinates run to 2):

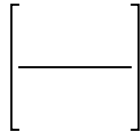
	<code>\matrixdiagram[3][2]{\mdFill[green!25]{0}{1}{0.5}{2}}</code>
	<code>\matrixdiagram[3][2]{\mdFill[blue!15]{0}{0}{1}{1}}</code>
	<code>\matrixdiagram[3][2]{\mdFill[gray!15]{0}{0}{0.75}{2}}</code>

5.2 `\mdVLine{x}` and `\mdHLine{y}`

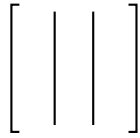
`\mdVLine{x}` runs from $(x, 0)$ to (x, H) ; `\mdHLine{y}` runs from $(0, y)$ to $(1, y)$.



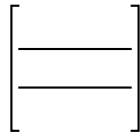
`\matrixdiagram[3]{\mdVLine{0.5}}`



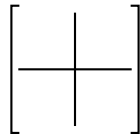
`\matrixdiagram[3]{\mdHLine{0.5}}`



`\matrixdiagram[3]{\mdVLine{0.33}\mdVLine{0.67}}`

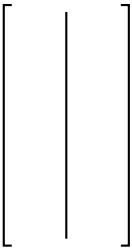


`\matrixdiagram[3]{\mdHLine{0.33}\mdHLine{0.67}}`

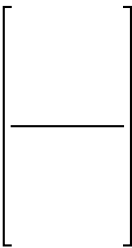


`\matrixdiagram[3]{\mdVLine{0.5}\mdHLine{0.5}}`

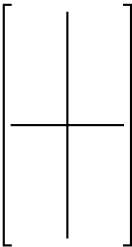
With `aspect = 2`:



`\matrixdiagram[3][2]{\mdVLine{0.5}}`

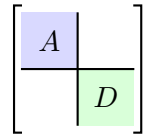
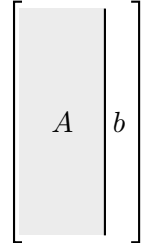


`\matrixdiagram[3][2]{\mdHLine{1.0}}`



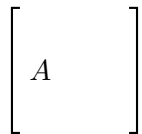
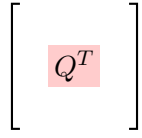
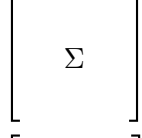
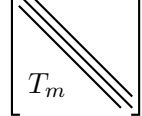
`\matrixdiagram[3][2]{\mdVLine{0.5}\mdHLine{1.0}}`

Combined with fills (block partitioning):

	<pre> \matrixdiagram[3]{ \mdFill[blue!15]{0}{0.5}{0.5}{1} \mdFill[green!15]{0.5}{0}{1}{0.5} \mdVLine{0.5}\mdHLine{0.5} \mdLabel{0.25}{0.75}{A} \mdLabel{0.75}{0.25}{D} } </pre>
	<pre> \matrixdiagram[3][2]{ \mdFill[gray!15]{0}{0}{0.75}{2} \mdVLine{0.75} \mdLabel{0.375}{1}{A} \mdLabel{0.875}{1}{b} } </pre>

6 Label commands

6.1 `\mdLabel{x}{y}{text}` — **math mode**

	<code>\matrixdiagram[3]{\mdLabel{0.2}{0.5}{A}}</code>
	<code>\matrixdiagram[3]{\mdLabel[red!20]{0.5}{0.5}{Q^T}}</code>
	<code>\matrixdiagram[3]{\mdLabel{0.5}{0.5}{\Sigma}}</code>
	<code>\matrixdiagram[3]{\mdDiagBand\mdLabel{0.25}{0.2}{T_m}}</code>

6.2 `\mdLabelCenter{text}` — **always centered**

Automatically placed at $(0.5, H/2)$ regardless of aspect ratio:

$\begin{bmatrix} Q \end{bmatrix}$	<code>\matrixdiagram[3][1]{\mdLabelCenter{Q}}</code>
$\begin{bmatrix} Q \end{bmatrix}$	<code>\matrixdiagram[3][2]{\mdLabelCenter{Q}}</code>
$\begin{bmatrix} Q \end{bmatrix}$	<code>\matrixdiagram[3][3]{\mdLabelCenter{Q}}</code>
$\begin{bmatrix} Q \end{bmatrix}$	<code>\matrixdiagram[3][0.5]{\mdLabelCenter{Q}}</code>

6.3 `\mdLabelRaw{x}{y}{text}` — raw text

$\begin{bmatrix} \text{dense} \end{bmatrix}$	<code>\matrixdiagram[3]{\mdLabelRaw{0.5}{0.5}{dense}}</code>
$\begin{bmatrix} \approx 0 \end{bmatrix}$	<code>\matrixdiagram[3]{\mdLabelRaw{0.5}{0.5}{\approx 0}}</code>

7 Spike commands

`\mdSpike{x}{y}{dir}` draws an L-shaped corner at (x, y) . The four directions connect to different pairs of edges:

$$\left[\begin{array}{c} \text{H} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixdiagram}[3]{\backslash\text{mdSpike}\{0.5\}\{0.5\}\{\text{tl}\}}$$

$$\left[\begin{array}{c} \text{L} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixdiagram}[3]{\backslash\text{mdSpike}\{0.5\}\{0.5\}\{\text{tr}\}}$$

$$\left[\begin{array}{c} \text{H} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixdiagram}[3]{\backslash\text{mdSpike}\{0.5\}\{0.5\}\{\text{bl}\}}$$

$$\left[\begin{array}{c} \text{L} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixdiagram}[3]{\backslash\text{mdSpike}\{0.5\}\{0.5\}\{\text{br}\}}$$

tl (top+left), tr (top+right), bl (bottom+left), br (bottom+right).

With fills:

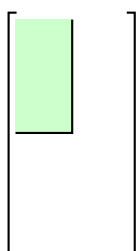
$$\left[\begin{array}{c} \text{Green} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixdiagram}[3]{\backslash\text{mdFill}[\text{green!20}]\{0\}\{0.5\}\{0.5\}\{1\}\backslash\text{mdSpike}\{0.5\}\{0.5\}\{\text{tl}\}}$$

$$\left[\begin{array}{c} \text{Blue} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixdiagram}[3]{\backslash\text{mdFill}[\text{blue!15}]\{0.5\}\{0.5\}\{1\}\{1\}\backslash\text{mdSpike}\{0.5\}\{0.5\}\{\text{tr}\}}$$

$$\left[\begin{array}{c} \text{Red} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixdiagram}[3]{\backslash\text{mdFill}[\text{red!15}]\{0\}\{0\}\{0.5\}\{0.5\}\backslash\text{mdSpike}\{0.5\}\{0.5\}\{\text{bl}\}}$$

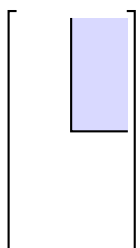
$$\left[\begin{array}{c} \text{Orange} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixdiagram}[3]{\backslash\text{mdFill}[\text{orange!20}]\{0.5\}\{0\}\{1\}\{0.5\}\backslash\text{mdSpike}\{0.5\}\{0.5\}\{\text{br}\}}$$

With aspect = 2, corner at mid-height (0.5, 1):



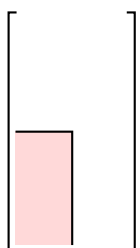
$$\begin{bmatrix} \text{green square} & \\ & \\ & \end{bmatrix}$$

`\matrixdiagram[3][2]{\mdFill[green!20]{0}{1}{0.5}{2}\mdSpike{0.5}{1}{t1}}`



$$\begin{bmatrix} & \text{blue square} \\ & \\ & \end{bmatrix}$$

`\matrixdiagram[3][2]{\mdFill[blue!15]{0.5}{1}{1}{2}\mdSpike{0.5}{1}{tr}}`



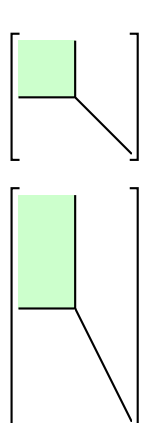
$$\begin{bmatrix} & \\ \text{red square} & \\ & \end{bmatrix}$$

`\matrixdiagram[3][2]{\mdFill[red!15]{0}{0}{0.5}{1}\mdSpike{0.5}{1}{bl}}`

8 Clipped diagonal commands

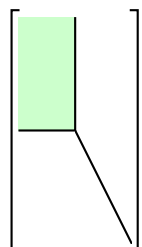
The ...From variants force the diagonal band from $[x_1, y_1]$ to $[x_2, y_2]$. Always pair with `\mdSpike` at the same corner.

8.1 `\mdDiagFrom{x1}{y1}{x2}{y2}`



$$\begin{bmatrix} \text{green square} & & \\ & & \\ & & \end{bmatrix}$$

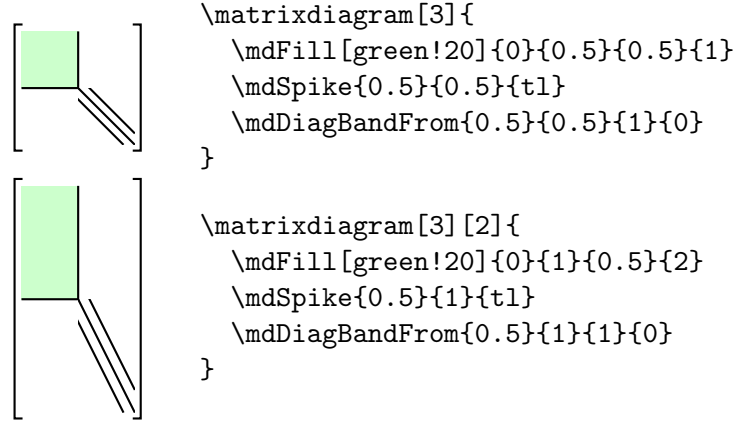
```
\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{t1}
  \mdDiagFrom{0.5}{0.5}{1}{0}
}
```



$$\begin{bmatrix} \text{green square} & & \\ & & \\ & & \end{bmatrix}$$

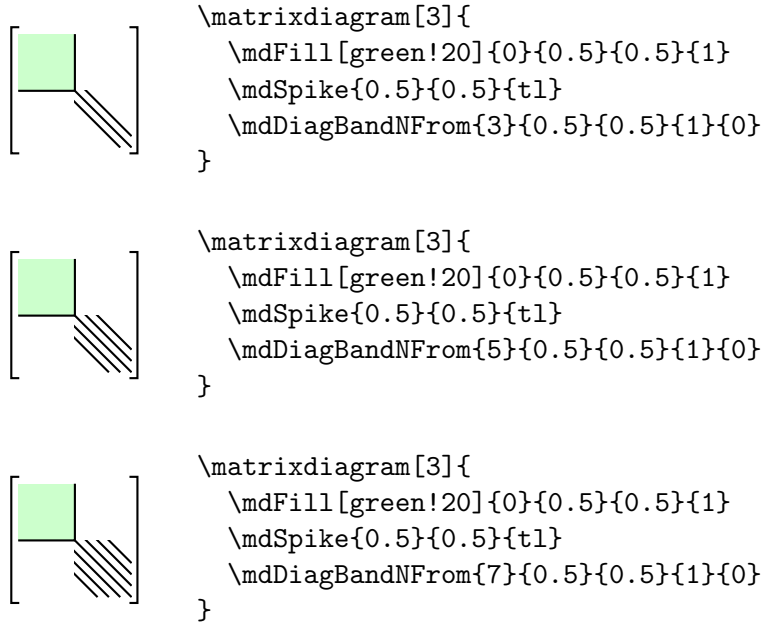
```
\matrixdiagram[3][2]{
  \mdFill[green!20]{0}{1}{0.5}{2}
  \mdSpike{0.5}{1}{t1}
  \mdDiagFrom{0.5}{1}{1}{0}
}
```

8.2 `\mdDiagBandFrom{cx}{cy}`



8.3 `\mdDiagBandNFrom{n}{cx}{cy}`

$n = 3, 5, 7$:



9 Realistic equations

9.1 QR decomposition: $A = QR$

$$\begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} Q \end{bmatrix} \begin{bmatrix} R \end{bmatrix}$$

```

\matrixdiagram[3][2]{\mdLabelCenter{A}} =
\matrixdiagram[3][2]{\mdLabelCenter{Q}}
\matrixdiagram[3]{\mdLabelCenter{R}}

```

9.2 Bidiagonalization: $A = UBV^T$

$$\begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} U \end{bmatrix} \begin{bmatrix} \diagdown \diagup \end{bmatrix} \begin{bmatrix} V^T \end{bmatrix}$$

```
\matrixdiagram[3]{\mdLabelCenter{A}} =
\matrixdiagram[3]{\mdLabelCenter{U}}
\matrixdiagram[3]{\mdBidiagUpper}
\matrixdiagram[3]{\mdLabelCenter{V^T}}
```

9.3 Similarity: $Q^T T_m Q = \tilde{T}_m$

$$\begin{bmatrix} Q^T \end{bmatrix} \begin{bmatrix} \diagdown \diagup \end{bmatrix} \begin{bmatrix} Q \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} \text{green square} \end{bmatrix} \\ \tilde{T}_m \end{bmatrix}$$

```
\matrixdiagram[3]{\mdLabelCenter{Q^T}}
\matrixdiagram[3]{
\mdDiagBand\mdLabel{0.25}{0.15}{T_m}
}
\matrixdiagram[3]{\mdLabelCenter{Q}} =
\matrixdiagram[3]{
\mdFill[green!25]{0}{0.5}{0.5}{1}
\mdSpike{0.5}{0.5}{t1}
\mdDiagBandFrom{0.5}{0.5}{1}{0}
\mdLabel{0.3}{0.12}{\widetilde{T}_m}
}
```

9.4 Hessenberg reduction: $Q^T A Q = H$

$$\begin{bmatrix} Q^T \end{bmatrix} \begin{bmatrix} A \end{bmatrix} \begin{bmatrix} Q \end{bmatrix} = \begin{bmatrix} \diagdown \diagup H \end{bmatrix}$$

```
\matrixdiagram[3]{\mdLabelCenter{Q^T}}
\matrixdiagram[3]{\mdLabelCenter{A}}
\matrixdiagram[3]{\mdLabelCenter{Q}} =
\matrixdiagram[3]{
\mdHessenberg\mdLabel{0.7}{0.55}{H}
}
```

9.5 Column-bordered decomposition: $AP_m = P_m T_m + f$

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} P_m \end{bmatrix} = \begin{bmatrix} P_m \end{bmatrix} \begin{bmatrix} \diagdown \diagup \end{bmatrix} + \begin{bmatrix} \text{gray square} \mid f \end{bmatrix}$$

```
\matrixdiagram[4]{\mdLabelCenter{A}}
\matrixdiagram[2]{\mdLabelCenter{P_m}} =
\matrixdiagram[2]{\mdLabelCenter{P_m}}
\matrixdiagram[2]{
\mdDiagBand \mdLabel{0.25}{0.15}{T_m}
} +
\matrixdiagram[2]{
\mdFill[gray!15]{0}{0}{0.75}{1}
\mdVLine{0.75} \mdLabel{0.875}{0.5}{f}
}
```

9.6 Schur complement structure

$$\begin{bmatrix} A & B \\ C & S \end{bmatrix}$$

```
\matrixdiagram[4]{
\mdFill[blue!10]{0}{0.5}{0.5}{1} \mdFill[green!10]{0.5}{0}{1}{0.5}
\mdFill[gray!10]{0.5}{0.5}{1}{1}
\mdVLine{0.5} \mdHLine{0.5}
\mdLabel{0.25}{0.75}{A} \mdLabel{0.75}{0.75}{B}
\mdLabel{0.25}{0.25}{C} \mdLabel{0.75}{0.25}{S}}
```

9.7 Bordered similarity: $\overline{Q}_k^T D_{k+1} \overline{Q}_k = \overline{T}_{k+1}$

$$\begin{bmatrix} \overline{Q}_k^T \\ 1 \end{bmatrix} \begin{bmatrix} D_{k+1} \\ \hline \hline \end{bmatrix} \begin{bmatrix} \overline{Q}_k \\ 1 \end{bmatrix} = \begin{bmatrix} \overline{T}_{k+1} \end{bmatrix}$$

```
\matrixdiagram[3]{
\mdFill[magenta!10]{0}{0.15}{0.85}{1}
\mdLabel{0.5}{0.5}{\overline{Q}_{k}^T}
\mdLabel{0.93}{0.08}{1}
}
\matrixdiagram[3]{
\mdDiag
\draw[thick] (0,0) -- (1,0) -- (1,\mdH);
\mdLabel{0.25}{0.45}{D_{k+1}}
}
\matrixdiagram[3]{
\mdFill[magenta!10]{0}{0.15}{0.85}{1}
\mdLabel{0.5}{0.5}{\overline{Q}_{k}}
\mdLabel{0.93}{0.08}{1}
} =
\matrixdiagram[3]{
\mdDiagBand
\mdLabel{0.25}{0.15}{\overline{T}_{k+1}}
}
```

10 Command reference

<code>\matrixdiagram[s][a]{c}</code>	Main command. <code>s</code> = scale (def. 1), <code>a</code> = aspect (def. 1 = square).
<code>\mdDiag</code>	Anti-diagonal $(0, H) \rightarrow (1, 0)$.
<code>\mdDiagBand</code>	Tridiagonal band, spacing $d = 0.1$.
<code>\mdDiagBandN{n}</code>	n -diagonal band (n odd), fixed step 0.1.
<code>\mdBidiagUpper</code>	Main diagonal + super-diagonal ($d = 0.1$).
<code>\mdBidiagLower</code>	Main diagonal + sub-diagonal ($d = 0.1$).
<code>\mdHessenberg</code>	Upper-triangular outline + sub-diagonal.
<code>\mdHessenberg[color]</code>	Same with filled interior.
<code>\mdDiagFrom{x1}{y1}{x2}{y2}</code>	Main diagonal, from (x_1, y_1) to (x_2, y_2)
<code>\mdDiagBandFrom{x1}{y1}{x2}{y2}</code>	Tridiagonal band, clipped.
<code>\mdDiagBandNFrom{n}{x1}{y1}{x2}{y2}</code>	n -diagonal band, clipped.
<code>\mdFill[col]{x0}{y0}{x1}{y1}</code>	Filled rectangle. Default color <code>gray!20</code> .
<code>\mdVLine{x}</code>	Vertical line, full height 0 to H .
<code>\mdHLine{y}</code>	Horizontal line, full width.
<code>\mdLabel{x}{y}{text}</code>	Math-mode label at (x, y) .
<code>\mdLabel[color]{x}{y}{text}</code>	Same with filled node label
<code>\mdLabelRaw{x}{y}{text}</code>	Raw-text label at (x, y) .
<code>\mdLabelCenter{text}</code>	Math label at center $(0.5, H/2)$, any aspect.
<code>\mdSpike{x}{y}{dir}</code>	L-corner; <code>dir</code> $\in \{\text{tl}, \text{tr}, \text{bl}, \text{br}\}$.
<code>\mdBlank</code>	Empty content (brackets only).
<code>\mdH</code>	Expands to current aspect value inside content.